

Math Made Clear

Optional Subtitle

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About the Author

Short author bio here. (e.g., background and inspiration; include website/contact if desired.) Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Acknowledgements

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Preface

Mathematics is a universal language, but learning it in a second language can be both a challenge and an opportunity. This book was created especially for Korean students who are studying math in English. Our purpose is to bridge the gap between two worlds—helping you master essential math concepts while also building your confidence in English.

The motivation for this book comes from the experiences of many students who have struggled to understand math vocabulary, instructions, and problem-solving strategies in English. We know that with the right support, every student can succeed. By providing clear explanations, bilingual vocabulary, and step-by-step examples, we hope to make math accessible and enjoyable for you.

Whether you are preparing for international exams, planning to study abroad, or simply want to strengthen your math and language skills, this book is here to support your journey. Remember, every great mathematician started with curiosity and perseverance. We hope this book inspires you to explore, ask questions, and discover the beauty of mathematics in both Korean and English.

Introduction

Welcome to this math book designed especially for Korean students learning English! This book is organized to help you build a strong foundation in mathematics while improving your English skills. Each section covers a key topic, starting with clear explanations and followed by examples, practice questions, and helpful tips.

The book is divided into major sections such as Numbers & Operations, Expressions & Equations, Geometry, Data & Statistics, and more. Within each section, you will find step-by-step guides, vocabulary lists, and real-world applications. Special features include bilingual glossaries, common mistakes to avoid, and mini quizzes to check your understanding.

To get the most out of this book, we recommend reading each section carefully, trying the examples on your own, and using the vocabulary lists to strengthen your math English. Do not hesitate to revisit earlier sections if you need a refresher. Remember, learning math in a new language takes time and practice—be patient with

yourself and celebrate your progress!

We hope this book becomes a valuable companion on your journey to mastering math and English together.

Chapter 1

Numbers & Operations

In this chapter, you will explore the world of numbers and how we use them every day. You will learn about place value, decimals, fractions, and more. Each section includes explanations, examples, and practice problems to help you master these important math skills.

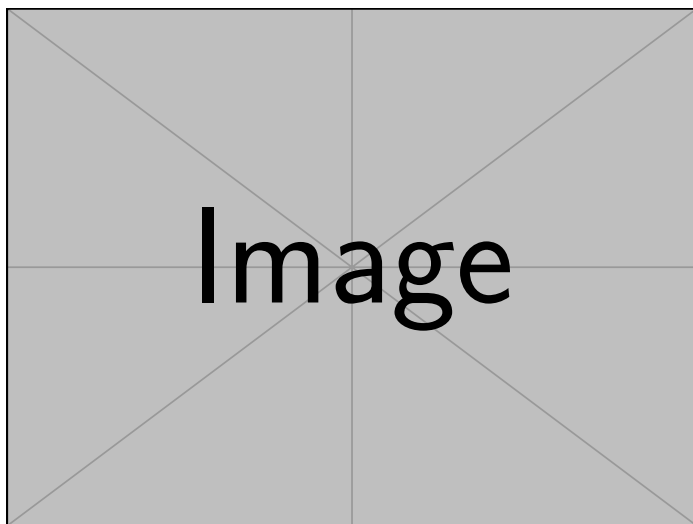


Figure 1.1: A scenic travel destination (example image).

1.1 Place Value

In our number system, every digit has a purpose. A 4 does not always mean 4. Sometimes it means 4000. That depends on where it sits in the number. This idea is called place value. Each place in a number is ten times the value of the one to its right. A number like 4275 is not just four digits. It is a structure: $4000 + 200 + 70 + 5$. We do not read numbers one digit at a time. We read them as a system, where each part works because of its position.

Place value turns a few digits into large or small numbers. Take 9836. The 8 is not just an 8. It is 800, because it sits in hundreds of places. In 5104, the 0 adds nothing but still matters. It shows that there are no tens. In 12345, the 2 sits in the thousands place and means 2000. These examples may seem simple, but they show a key truth: digits mean more when you know where they stand.

Place value is more than vocabulary. It affects how we add, subtract, estimate, and think about numbers. When we know what each digit means, we can break numbers apart and put them back together. We stop guessing and start understanding. That shift helps us see math not as a list of rules but as a system that makes sense.

Place value tells us the value of each digit in a number, depending on its position. In our decimal system, each place is ten times the value of the place to its right.

Example: Understanding Place Value

To truly grasp the concept of place value, let's take a closer look at the number 4275. Here, the digit 4 is found in the thousands place, which means it actually represents 4000, not just 4. Moving to the right, the 2 is in the hundreds place, so it stands for 200. The 7, sitting in the tens place, is worth 70, and finally, the 5 in the ones place simply means 5. This system, where each digit's value increases by a factor of ten as you move left, is what makes our decimal system so powerful and flexible.

To help visualize this, consider the following place value chart:

Place	Value
Thousands	1000
Hundreds	100
Tens	10
Ones	1

Let's see how this works in practice. If we write 4275 in expanded form, we get:

$$4275 = 4000 + 200 + 70 + 5$$

Now, imagine you are looking at the number 9836. You might wonder, what is the value of the digit 8 in this number? Since the 8 is in the hundreds place, its value is 800. This simple observation shows how the position of a digit determines its true value. Similarly, if you break down 5104 into its expanded form, you see it as $5000 + 100 + 0 + 4$, making it clear what each digit stands for. In another example, the digit 2 in 12345 is in the thousands place, so it represents 2000. Understanding these positions helps us see the true value of every digit in a number, which is a fundamental skill in mathematics.

1.1.1 Practice Questions

1. Short Question

What is the value of the digit 8 in the number 9836?

Answer: The digit 8 is in the **hundreds place**, so its value is **800**.

2. Long Question

Write the number 5,041 in expanded form. Then explain what each digit means based on its place.

Answer:

Expanded form: $5000 + 0 + 40 + 1$

Explanation:

- $5 = 5000$ (thousands place)
- $0 = 0$ (hundreds place)
- $4 = 40$ (tens place)
- $1 = 1$ (ones place)

3. Challenge Question

Two numbers have the same digits but in a different order: 5,104 and 4,105. Which number is greater, and how do you know? Explain using place value.

Answer:

5,104 is greater because the **5** is in the **thousands place**, meaning 5000.

In 4,105, the **4** is in the thousands place, meaning 4000.

So: **5,104 > 4,105**

1.1.2 Q&A 1 (English)

Short Question: Value of a Digit

What is the value of the digit 8 in the number 9836?

1.1.3 Q&A 1 (Korean)**Short Question: Value of a Digit**

What is the value of the digit 8 in the number 9836?

1.1.4 Q&A 1 (Typed Answer)

Short Question: Value of a Digit

What is the value of the digit 8 in the number 9836?

1.1.5 Q&A 2 (English)

Long Question: Expanded Form

Write the number 5,041 in expanded form. Then explain what each digit means based on its place.

1.1.6 Q&A 2 (Korean)

Long Question: Expanded Form

Write the number 5,041 in expanded form. Then explain what each digit means based on its place.

1.1.7 Q&A 2 (Typed Answer)

Long Question: Expanded Form

Write the number 5,041 in expanded form. Then explain what each digit means based on its place.

1.1.8 Q&A 3 (English)

Challenge Question: Comparing Numbers

Two numbers have the same digits but in a different order: 5,104 and 4,105. Which number is greater, and how do you know? Explain using place value.

1.1.9 Q&A 3 (Korean)

Challenge Question: Comparing Numbers

Two numbers have the same digits but in a different order: 5,104 and 4,105. Which number is greater, and how do you know? Explain using place value.

1.1.10 Q&A 3 (Typed Answer)

Challenge Question: Comparing Numbers

Two numbers have the same digits but in a different order: 5,104 and 4,105. Which number is greater, and how do you know? Explain using place value.

1.1.11 Hand-drawing 1

This is where you will see a hand-drawn answer to the previous question in **English**.

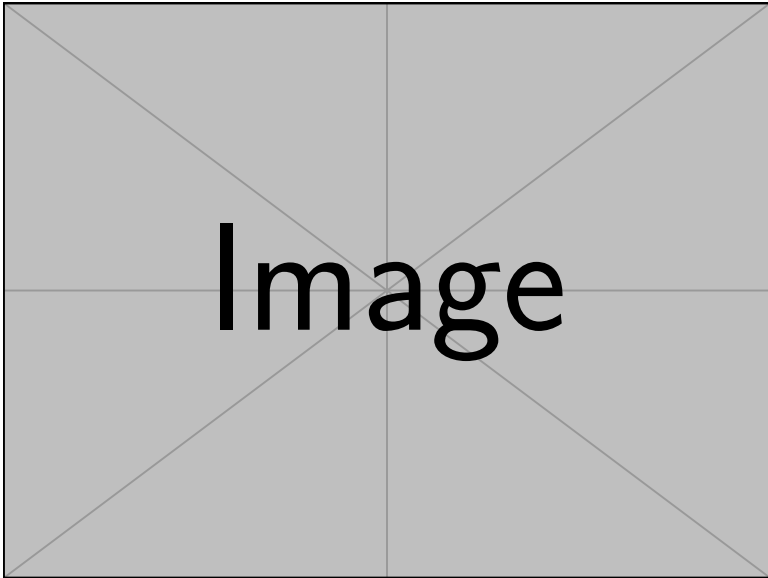


Figure 1.2: A hand-drawn answer in English.

1.1.12 Hand-drawing 2

This is where you will see a hand-drawn answer to the previous question in **Korean**.

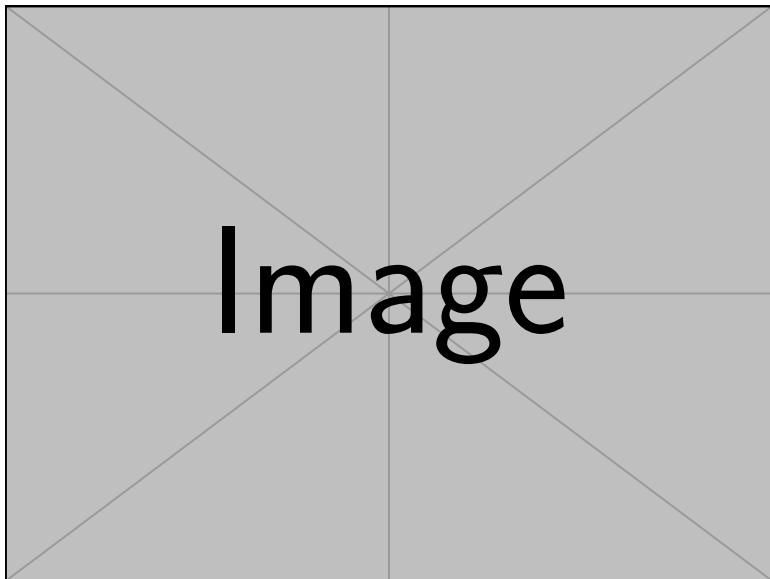


Figure 1.3: A hand-drawn answer in Korean.

1.2 Decimals and Fractions

Sometimes, a whole number is too much. We want to describe something in parts: half a sandwich, a third of a mile, seventy-five hundredths of a dollar. That's when we use fractions and decimals. There are two ways of saying the same thing: less than one. Fractions use numerators and denominators. Decimals use place value. But both express parts of a whole.

Decimals and fractions are not separate languages. They translate. 0.75 is the same as $75/100$, which reduces to $3/4$. The fraction $3/10$ becomes 0.3. The decimal 0.6 becomes $6/10$, which is $3/5$. Being able to switch between them is not just a trick. It is a skill. And that skill helps you compare, calculate, and make better sense of the numbers you see every day.

“Decimals and fractions are not separate languages.”

Example: Decimals and Fractions Conversion

As we move beyond whole numbers, we encounter quantities that are less than one. These are often represented as fractions or decimals. A **fraction** shows a part of a whole, such as $\frac{1}{2}$ or $\frac{3}{4}$. On the other hand, a **decimal** expresses the same idea using a decimal point, like 0.5 or 0.75. Both forms are just different ways to describe the same concept.

For example, consider the decimal 0.75. This can be written as the fraction $\frac{75}{100}$, which simplifies to $\frac{3}{4}$:

$$0.75 = \frac{75}{100} = \frac{3}{4}$$

Let's try converting between these forms. If you have the fraction $\frac{3}{10}$, you can write it as the decimal 0.3. Conversely, if you start with the decimal 0.6, you can express it as a fraction: $0.6 = \frac{6}{10} = \frac{3}{5}$ in simplest form. Being able to move between decimals and fractions is a key skill that will help you in many areas of math.

1.2.1 Practice Questions

1. **Short Question**

Write the fraction $\frac{3}{10}$ as a decimal.

Answer: 0.3 — Divide 3 by 10 to get the decimal.

2. **Long Question**

Convert the decimal 0.6 into a simplified fraction. Then explain each step.

Answer:

Step 1: Write 0.6 as $\frac{6}{10}$

Step 2: Simplify by dividing top and bottom by 2: $\frac{6}{10} = \frac{3}{5}$

Answer: $\frac{3}{5}$

3. **Challenge Question**

Emma says that the fraction $\frac{4}{5}$ is equal to the decimal 0.45. Is she correct? Explain your reasoning.

Answer:

No, because $\frac{4}{5} = 0.8$, not 0.45.

$0.45 = \frac{9}{20}$, not the same as $\frac{4}{5}$.

1.2.2 Hand-drawing 1

This is where you will see a hand-drawn answer to the previous question in **English**.

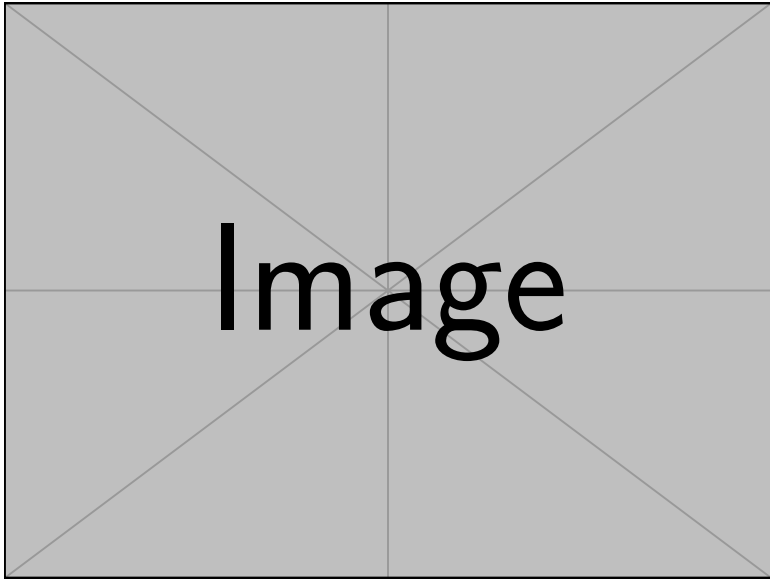


Figure 1.4: A hand-drawn answer in English.

1.2.3 Hand-drawing 2

This is where you will see a hand-drawn answer to the previous question in **Korean**.

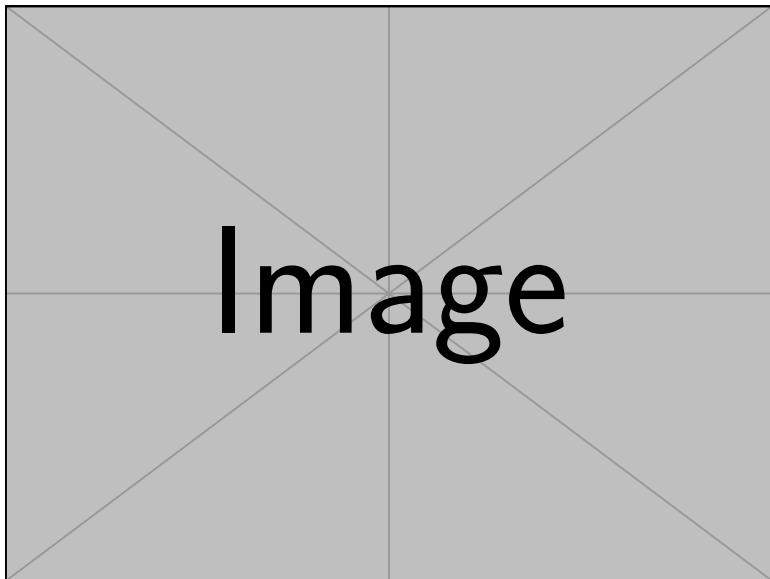


Figure 1.5: A hand-drawn answer in Korean.

1.3 Comparing and Ordering Numbers

Math is not just about knowing what numbers are. It is about knowing how they relate. We compare. We rank. We ask, which is bigger? Which is smaller? Which is equal? Symbols like $>$, $<$, and $=$ help us answer. But first, we have to understand the values.

Take 5.03 and 5.3. They may look close. But 5.3 is larger. Now try comparing fractions: $\frac{2}{5}$ and $\frac{3}{7}$. Convert them to decimals. $\frac{2}{5}$ is 0.4. $\frac{3}{7}$ is about 0.428. So $\frac{3}{7}$ is greater. This kind of comparison matters in real life—from prices to measurements to data.

Ordering numbers builds the same skill. Imagine you are given 0.2, $\frac{1}{3}$, 0.25, and 0.199. Which is the greatest? Which is least? Convert, compare, and you find the order. This process teaches more than math. It teaches precision, logic, and clarity.

Example: Comparing and Ordering Numbers

In mathematics, it's important not only to understand numbers, but also to compare them and put them in order. We use symbols like $>$ (greater than), $<$ (less than), and $=$ (equal to) to show how numbers relate to each other. For example, if you are given two numbers, you can decide which is larger or smaller, or if they are the same.

Let's look at a few examples. Suppose you want to compare 5.03 and 5.3. Since $5.03 < 5.3$, we use the less than symbol. Now, what if you have two fractions, $\frac{2}{5}$ and $\frac{3}{7}$? By converting them to decimals, you find that $\frac{2}{5} = 0.4$ and $\frac{3}{7} \approx 0.428$, so $0.4 < 0.428$, which means $\frac{2}{5} < \frac{3}{7}$.

Ordering numbers is also a useful skill. Imagine you are asked to arrange the numbers 0.2, $\frac{1}{3}$, 0.25, and 0.199 from greatest to least. By converting them all to decimals, you see that $\frac{1}{3} (\approx 0.333)$ is the largest, followed by 0.25, then 0.2, and finally 0.199. Sometimes, you may need to compare a fraction and a decimal, such as $\frac{4}{9}$ and 0.44. Since $\frac{4}{9} \approx 0.444$, it is slightly greater than 0.44. These skills are essential for working with numbers in real-life situations.

1.3.1 Practice Questions

1. **Short Question**

Which is greater: 5.03 or 5.3?

Answer: 5.3 is greater — it has more value in the tenths place.

2. **Long Question**

Put the following numbers in order from least to greatest:
0.199, $\frac{1}{3}$, 0.25, 0.2

Answer:

Convert to decimals:

- 0.199

- $\frac{1}{3} \approx 0.333$

- 0.25

- 0.2

Ordered: 0.199 ; 0.2 ; 0.25 ; $\frac{1}{3}$

3. **Challenge Question**

Which is greater: $\frac{4}{9}$ or 0.44? Show your reasoning.

Answer:

$\frac{4}{9} \approx 0.444...$ which is slightly greater than 0.44.

Answer: $\frac{4}{9}$ is greater.

1.3.2 Hand-drawing 1

This is where you will see a hand-drawn answer to the previous question in **English**.

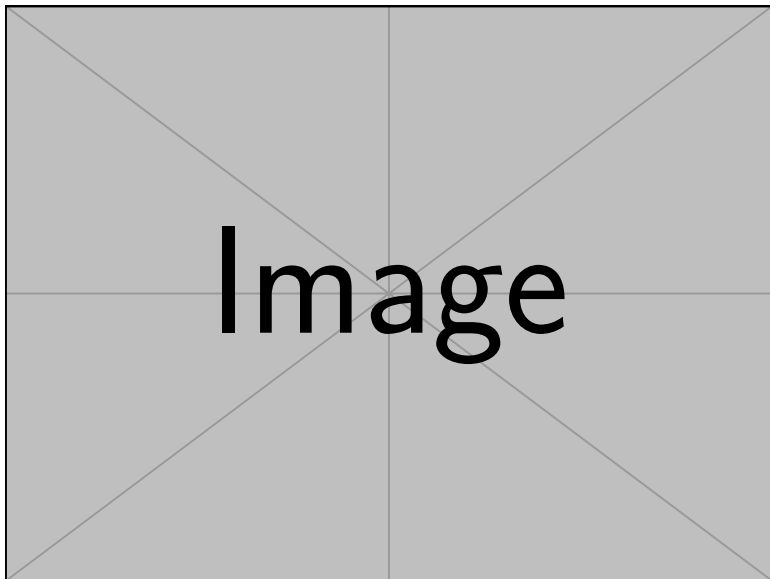


Figure 1.6: A hand-drawn answer in English.

1.3.3 Hand-drawing 2

This is where you will see a hand-drawn answer to the previous question in **Korean**.

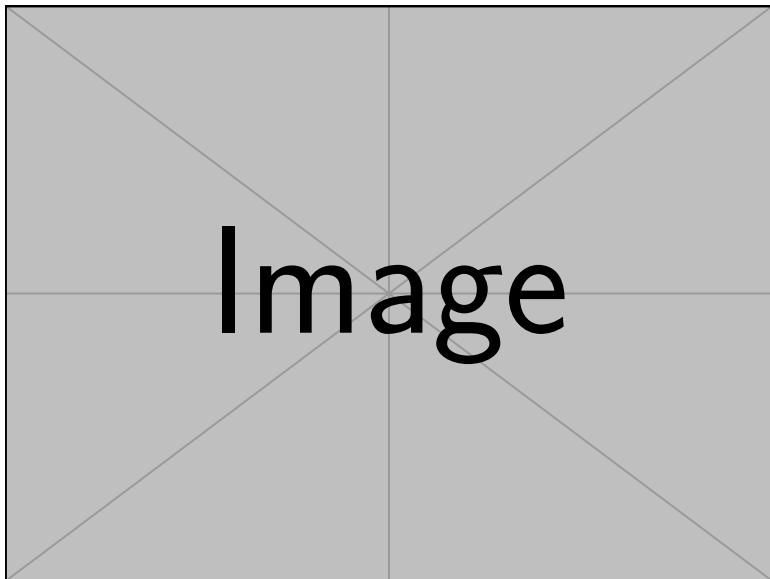


Figure 1.7: A hand-drawn answer in Korean.

1.4 Factors and Multiples

Some numbers fit together cleanly. Others do not. Factors and multiples explain why. A factor is a number that divides evenly. A multiple is what you get when you multiply. These two ideas help us break numbers apart and build them back up.

If you list the factors of 12, you get 1, 2, 3, 4, 6, and 12. Each divides 12 without a remainder. The multiples of 5 are 5, 10, 15, 20, and so on. These ideas may seem mechanical, but they are powerful. They lay the groundwork for division, fractions, and algebra.

1.4.1 Example: Factors and Multiples

As you continue your journey in mathematics, you'll often need to break numbers down or build them up. This is where factors and multiples come in. A **factor** is a number that divides another number exactly, leaving no remainder. For example, the factors of 12 are 1, 2, 3, 4, 6, and 12, because each of these divides 12 evenly. On the other hand, a **multiple** is what you get when you multiply a number by an integer. For instance, the multiples of 5 are 5, 10, 15, 20, and so on.

Let's see how this works with a few examples. If you want to find all the factors of 24, you list out 1, 2, 3, 4, 6, 8, 12, and 24. If you're asked for the first six multiples of 8, you simply multiply: 8, 16, 24, 32, 40, and 48. Sometimes, you might wonder if one number is a factor of another. For example, is 9 a factor of 81? Since $81 \div 9 = 9$ with no remainder, the answer is yes. Understanding factors and multiples is crucial for topics like division, fractions, and finding patterns in numbers.

1.4.2 Practice Questions

1. **Short Question**

List all the factors of 18.

Answer: 1, 2, 3, 6, 9, 18

2. **Long Question**

What are the first six multiples of 7? Explain how you found them.

Answer:

7, 14, 21, 28, 35, 42 — Multiply 7 by 1 through 6.

3. **Challenge Question**

Is 9 a factor of 81? Is 81 a multiple of 9? Explain.

Answer:

Yes. $81 \div 9 = 9$ — so 9 is a factor, and 81 is a multiple.

1.4.3 Hand-drawing 1

This is where you will see a hand-drawn answer to the previous question in **English**.

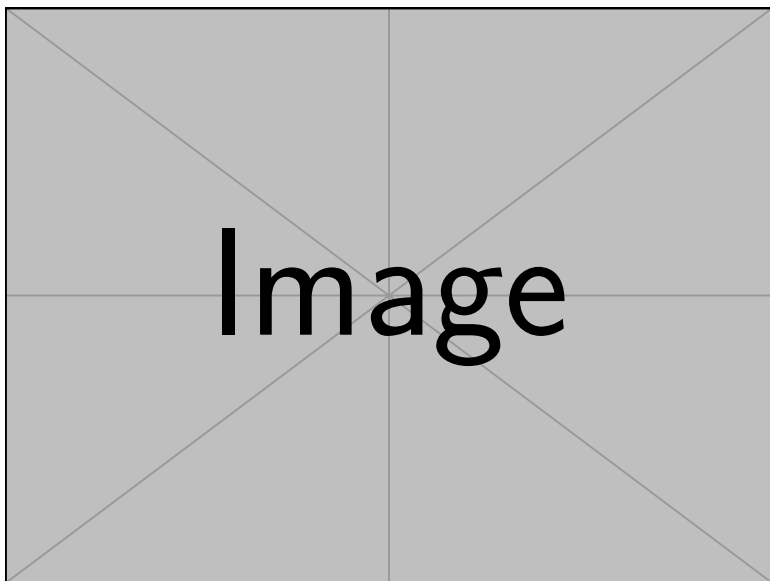


Figure 1.8: A hand-drawn answer in English.

1.4.4 Hand-drawing 2

This is where you will see a hand-drawn answer to the previous question in **Korean**.

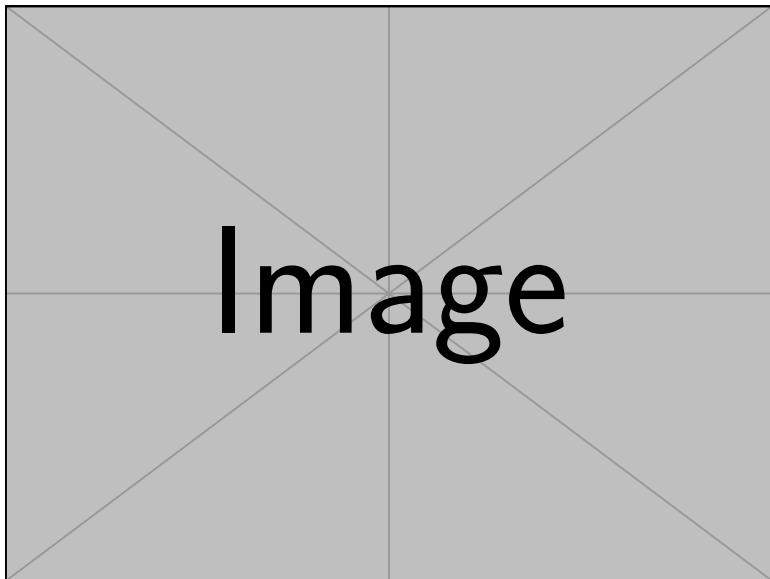


Figure 1.9: A hand-drawn answer in Korean.

1.5 Prime and Composite Numbers

Some numbers stand alone. Others are made of parts. Prime numbers have only two factors: 1 and themselves. Composite numbers have more. This is not just a label—it is a clue to how numbers work.

2, 3, 5, 7, 11, 13, and 17 are prime. They cannot be broken down. But 15 is composite: it splits into 3 and 5. Learning to recognize prime and composite numbers helps us factor numbers, simplify fractions, and understand the building blocks of math.

1.5.1 Example: Prime and Composite Numbers

As you explore numbers further, you'll discover that some numbers have only two factors, while others have many. A **prime number** is a number greater than 1 that has exactly two distinct factors: 1 and itself. For example, 2, 3, 5, 7, 11, 13, 17, and 19 are all prime numbers under 20. In contrast, a **composite number** has more than two factors. For instance, 15 is composite because it can be divided by 1, 3, 5, and 15.

Let's practice identifying these. Consider the numbers 14, 17, 21, 23, and 25. Among these, 17 and 23 are prime because they have only two factors. The others—14, 21, and 25—are composite. If you look at all the composite numbers between 10 and 20, you'll find 10, 12, 14, 15, 16, 18, and 20. It's also important to note that the number 1 is neither prime nor composite, since it has only one factor: itself. Recognizing prime and composite numbers is a foundational skill for understanding more advanced math concepts like prime factorization and divisibility.

1.5.2 Practice Questions

1. **Short Question**

Is 11 a prime or composite number?

Answer: Prime — only two factors: 1 and 11.

2. **Long Question**

Identify all the prime numbers between 10 and 20. Explain how you know.

Answer:

11, 13, 17, 19 — they each have exactly two factors: 1 and themselves.

3. **Challenge Question**

Why is the number 1 neither prime nor composite?

Answer:

It has only **one factor** (itself), not two or more. So it's **neither**.

1.5.3 Prime Factorization Questions

1. **Prime Factorization (Multiple Choice)**

What is the prime factorization of 60?

A. $2 \times 2 \times 3 \times 5$

B. $2 \times 3 \times 10$

C. $2 \times 5 \times 6$

D. $3 \times 4 \times 5$

Answer: The correct answer is **A**. The prime factorization of 60 is $2 \times 2 \times 3 \times 5$ (or $2^2 \times 3 \times 5$).

2. **Prime Factorization (Open-Ended)**

Find the prime factorization of 84.

Answer: The prime factorization of 84 is $2 \times 2 \times 3 \times 7$ (or $2^2 \times 3 \times 7$).

3. **Prime Factorization (Open-Ended)**

Write the prime factorization of 105 using exponents.

Answer: The prime factorization of 105 is $3 \times 5 \times 7$. Since all are single primes, no exponents are needed.

4. **Prime Factorization (Multiple Choice)**

Which of the following is the prime factorization of 72?

A. $2^3 \times 3^2$

B. $2^2 \times 3^3$

C. $2^4 \times 3$

D. 2×6^2

Answer: The correct answer is **A**. $72 = 2^3 \times 3^2$.

1.5.4 Hand-drawing 1

This is where you will see a hand-drawn answer to the previous question in **English**.

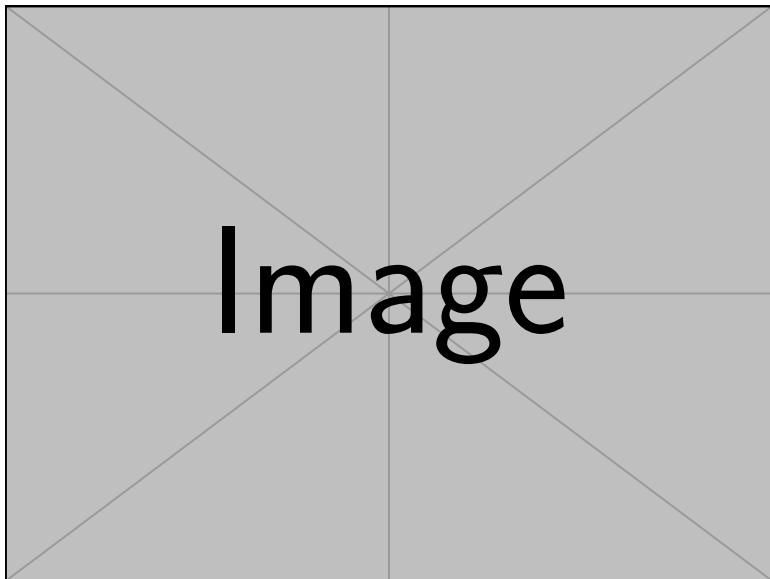


Figure 1.10: A hand-drawn answer in English.

1.5.5 Hand-drawing 2

This is where you will see a hand-drawn answer to the previous question in **Korean**.

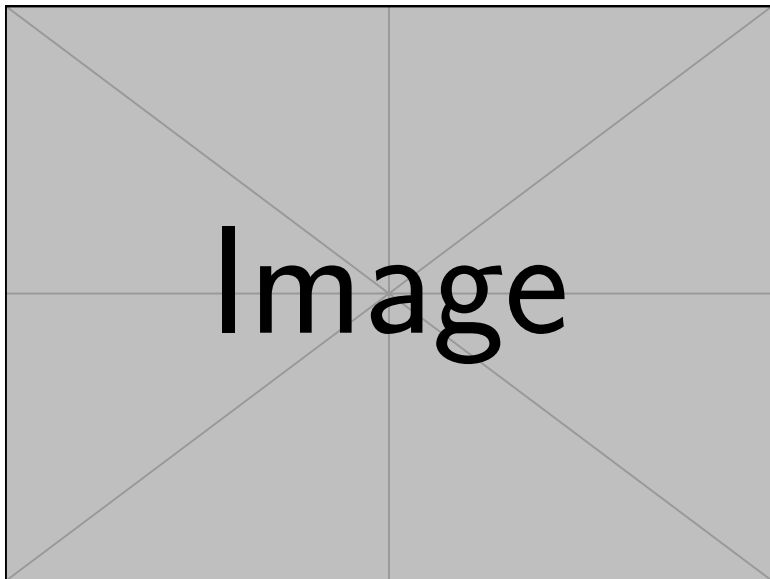


Figure 1.11: A hand-drawn answer in Korean.

1.6 Greatest Common Factor & Least Common Multiple

Sometimes two numbers meet in the middle. The Greatest Common Factor (GCF) is the biggest number they both share as a factor. The Least Common Multiple (LCM) is the smallest number they both share as a multiple. These ideas sound opposite, but they both help us solve problems with shared values.

Take 12 and 18. Their GCF is 6 since it is the biggest number that divides them both. Their LCM is 36 which is the smallest number they both fit into. We find these using prime factorization. Once you learn how, you see how numbers connect. And once you see the connection, math becomes clearer.

1.6.1 Example: GCF and LCM

When working with two or more numbers, it's often useful to find numbers they have in common. The **Greatest Common Factor (GCF)** is the largest number that divides two numbers exactly, while the **Least Common Multiple (LCM)** is the smallest number that both numbers divide into evenly.

To find the GCF and LCM, you can use prime factorization. For example, let's find the GCF and LCM of 12 and 18. The prime factorization of 12 is $2^2 \times 3$, and for 18 it's 2×3^2 . The GCF is the product of the lowest powers of common primes: $2^1 \times 3^1 = 6$. The LCM is the product of the highest powers: $2^2 \times 3^2 = 36$.

Let's apply this to a few more examples. The GCF of 24 and 36 is 12, since 12 is the largest number that divides both. The LCM of 8 and 12 is 24, which is the smallest number both can divide into. For 14 and 28, we see that $14 = 2 \times 7$ and $28 = 2^2 \times 7$, so the GCF is 14 and the LCM is 28. Mastering GCF and LCM will help you with fractions, ratios, and solving many real-world problems.

1.6.2 Hand-drawing 1

This is where you will see a hand-drawn answer to the previous question in **English**.

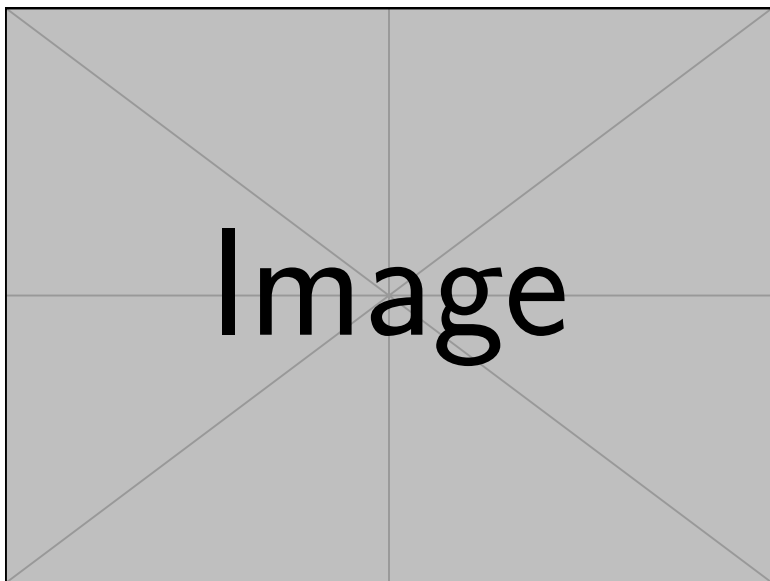


Figure 1.12: A hand-drawn answer in English.

1.6.3 Hand-drawing 2

This is where you will see a hand-drawn answer to the previous question in **Korean**.

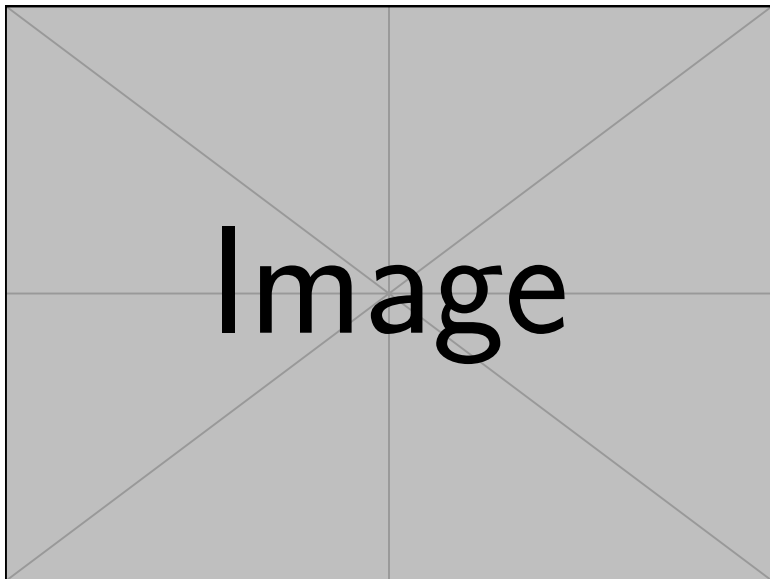


Figure 1.13: A hand-drawn answer in Korean.

1.7 Summary and Tips

Interesting Facts

- The word “digit” comes from the Latin word for finger, because people used their fingers to count.
- The decimal system is called “base-10” because each place is ten times the value of the place to its right.
- The largest single-digit number in the decimal system is 9.
- The introduction of zero by Indian mathematicians made calculations much clearer and easier.

Tip for Students

Math is not just about solving problems. It is about seeing how ideas fit together. Learn the basics—place value, fractions, comparisons, factors, primes, and multiples. They are not just chapters. They are tools. And the better you understand them, the more confidently you can use them anywhere.

Appendix

Important Formulas and Definitions

- **Place Value:** Value of a digit = digit $\times$ place value.
- **Fraction to Decimal:** $\frac{a}{b} = a \div b$.
- **Prime Number:** A number greater than 1 with exactly two factors: 1 and itself.
- **Composite Number:** A number with more than two factors.
- **GCF:** Product of lowest powers of common primes.
- **LCM:** Product of highest powers of all primes present.

Further Reading and Resources

- "Elementary Mathematics" by Liping Ma
- "How to Be a Math Genius" by Mike Goldsmith
- Khan Academy: <https://www.khanacademy.org/>
- Purplemath: <https://www.purplemath.com/>

Summary of Key Concepts

Topic	Key Points
Place Value	Each digit's value depends on its position. Example: $4275 = 4000 + 200 + 70 + 5$.
Decimals and Fractions	Decimals and fractions both represent parts of a whole. Example: $0.75 = \frac{3}{4}$.
Comparing and Ordering Numbers	Use $>$, $<$, $=$ to compare. Convert to decimals for easy comparison. Example: $\frac{2}{5} < \frac{3}{7}$.
Factors and Multiples	Factors divide a number evenly; multiples are products of a number and an integer. Example: Factors of 12 are 1, 2, 3, 4, 6, 12.
Prime and Composite Numbers	Primes have exactly two factors (1 and itself); composites have more. Example: 2, 3, 5 are prime; 4, 6, 8 are composite.
GCF and LCM	GCF: largest common factor; LCM: smallest common multiple. Use prime factorization. Example: GCF of 12 and 18 is 6; LCM is 36.

Table 1.1: **Summary of Key Concepts**

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